

REPORT 600 WORDS

CSC8501 - Prisoners Dilemma Oliver Wilde

Overview

I coded a self contained suite for the Iterated Prisoners Dilemma (IPD) with a single command line interface (CLI). It runs five studies as required by the brief: Q1-Q5, which are the baseline tournament with confidence intervals and a pairwise payoff matrix, a noise sensitive sweep, analytical robustness against exploiter strategies, and an evolutionary selection process, and lastly a strategic complexity budget (SCB).The program does include the standard strategies: ALLC, ALLD, TFT, GRIM, PAVLOV, PROBER, CTFT, RND, and two originals called TTFT and GRUDGE2. All of the experiments are reproducible using fixed seeds and a shared RNG.

Methods and configuration

Unless specifically stated otherwise, I used the Prisoner's Dilemma payoffs as 5,3,1,0 for T,R,P,S respectively with 150 rounds per match, with 30 repeats under fixed seeds. Noise ϵ flips a players intended move with a probability ϵ variable. The CLI will accept comma separated strategy lists and supports `--format text|csv|json` . For each strategy I reported a 95% confidence interval computed as $1.96 \times (\text{standard deviation} / \sqrt{\text{repeats}})$. The program does validate payoffs as well as verifying explicit override using `--force` (order matters: `--force` must precede `--payoffs`).

Q1 – Baseline tournament ($\epsilon = 0$)

Command (representative):

- `-run Q1 --rounds 150 --repeats 30 --epsilon 0 --seed 42 --payoffs 5,3,1,0 --strategies ALLC,ALLD,TFT,GRIM,PAVLOV,CTFT,PROBER,RND0.3`

Results: The leaderboard ranks TFT (~2592.13), PAVLOV (~2591.47), and CTFT (~2590.57) at the top then followed by PROBER, GRIM, RND(0.3), ALLC, and ALLD, all with tight CI95 values. The pairwise matrix shows mutual cooperators at roughly 450 per match (150 × R), and ALLD vs ALLC = 750 (150 × T). This confirms the expected pattern: in clean play, reciprocal strategies dominate by sustaining mutual cooperation.

Q2 – Noise sensitivity ($\epsilon \in \{0, 0.05, 0.10, 0.20\}$)

Command:

- `-run Q2 --epsilon 0.0,0.05,0.10,0.20 --strategies TFT,GRIM,PAVLOV,CTFT`

Results: At $\epsilon=0$ the four strategies tie at 1350 (full cooperation). With $\epsilon=0.05$, GRIM collapses (~651.5) while CTFT/TFT/PAVLOV stay near 900. At $\epsilon=0.10$, CTFT keeps a small edge. At $\epsilon=0.20$, the ordering compresses and can flip with GRIM at 937.7 so it's slightly ahead, highlighting that strong noise undermines long memory discipline and makes outcomes more stochastic. Overall: GRIM is brittle to accidental defections, while CTFT/TFT/PAVLOV are quite noise-tolerant.

Q3 – Robustness versus exploiters

Command:

- `-run Q3 --epsilon 0.05 --strategies ALLC,TFT,CTFT,PAVLOV,PROBER,ALLD`

Results: At $\epsilon=0$, PROBER leads (~2028) by exploiting predictable opponents. At $\epsilon=0.05$, CTFT rises (~1722) and PROBER's advantage shrinks. This shows that light noise reduces exploiter headroom, favouring forgiving/repairing strategies.

Q4 – Evolutionary selection

Command:

- `-run Q4 --evolve 1 --population 200 --generations 120 --mutation 0.02 --epsilon 0.05`

Results: The final population converges to a reciprocity heavy mix: **CTFT ~42.8%, PAVLOV ~17.5%, TFT ~15.3%, PROBER ~14.6%**, with small minorities of **ALLC/ALLD/GRIM**. This is consistent with Q1–Q3 under moderate noise:

robust reciprocators dominate, but opportunists still retain a small population in proportion.

Q5 – Strategic Complexity Budget (SCB)

Command:

- `-run Q5 --scb on --scb-cost ALLC=1,ALLD=1,TFT=2,GRIM=2,PAVLOV=2,CTFT=3,PROBER=3,TTFT=2,GRUDGE2=2 --epsilon 0.03`

Results: After normalising each strategy's score by $(1 + \text{cost})$, simpler policies move up the table (e.g., ALLC and ALLD gain relative position), while higher-cost strategies drop accordingly. My originals (TTFT, GRUDGE2) land mid table under this newly introduced budget. This demonstrates how conclusions change when model complexity is priced, as requested by the brief.

Reproducibility and validation

I showed reproducibility using `--save/--load` (store a config, then replay Q1 to obtain identical JSON and the same 2×2 pairwise matrix). The payoff validator rejects non PD tuples unless `--force` is provided before `--payoffs`. All randomness (noise and $\text{RND}(p)$) is driven by a single RNG seeded from the CLI, so runs repeat exactly with the same seed.

Conclusion

The implementation I provide delivers on the requirements for Q1-Q5 via a CLI with clear outputs. The results I obtained do match with established IPD expectations; cooperation wins without noise, moderate noise will penalise brittle punishment (GRIM) and exploiters (PROBER). CTFT/TFT/PAVLOV perform best in noisy play, evolution converges toward those robust strategies, and the SCB reorders the table toward simpler policies. The submitted outputs support this narrative and are reproducible from the provided commands.